# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Fourth Semester of B.Tech University Examination (CE)**
May 2014

**CE209: Design & Analysis of Algorithms**

**Date: 05.05.2014, Monday** | **Time: 10:00 am to 1:00 pm** | **Maximum Marks:70**

*Instructions:*
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION-I

**Q-1 Do as directed:**

(a) Check correctness of the following relationship: **01**
$2^{2n}=O(2^n)$

(b) For which value of n, n-queen problem has no solution? Give reasons. **01**

(c) What is the smaller value of N such that an algorithm whose running time is $100n^2$ runs faster than an algorithm whose running time is $2^n$ on the same machine? **02**

(d) Find out the time complexity of following program: **02**

```
bool duplicate=false;
for(int i=0;i<n;++i){
  for(int j=0;i<n;++j){
    if((i!=j && A[i]= =A[j])){
      duplicate=true;
      break;
    }
  }
if(duplicate){
break;
}
}
```

(d) The worst case time complexity of merge sort algorithm is ____________. **01**

**Q-2 Give the answer of the following questions.**

(a) Solve these equations using Master Method: **04**
1. $T(n)=2T(n/2)+n^3$
2. $T(n)=2T(n/4)+ \text{sqrt}(n)$

(b) Consider the following recurrence **05**
$T(n)=3T(n/4) + n$
Obtain the asymptotic bound for the above recurrence by recursion tree method.

(c) Define time complexity & Space Complexity. Explain with help of suitable example Best Case, Average Case and Worst Case Time Complexity of an algorithm. **05**

**OR**

**Q-2 Give the answer of the following questions.**

(a) Write down the algorithm for Insertion Sort (descending order). Also find time complexity of it. **04**

(b) Consider the recurrence following recurrence $T(n)=2T(n/2)+n^2$, Obtain the asymptotic bound **05** using recursion tree method.

(c) Write down an algorithm for DFS also derive its time complexity. Also compare DFS algorithm with BFS algorithm. 05

**Q-3 Attempt Any Two.** 14

(a) Use branch and bound to solve the assignment problem (Minimization) with the following cost matrices.

| | Job 1 | Job 2 | Job 3 | Job 4 |
|---|---|---|---|---|
| Person a | 94 | 1 | 54 | 68 |
| Person b | 74 | 10 | 88 | 82 |
| Person c | 62 | 88 | 8 | 76 |
| Person d | 11 | 74 | 81 | 21 |

(b) Mention Naïve string matching algorithm; also derive its time complexity. Show the comparisons the naïve string matcher makes for the pattern=0001 in the text T=000010001010001.

(c) How many valid match and spurious hits does the Rabin-Karp matcher encounter in the T=2359023141526739921 when looking for the pattern = 31415. Take value of modulo q=13. Show the complete procedure using neat figures.

## SECTION-II

**Q-4 Do as directed:**

(a) What is relationship between NP, P, NP Hard and NP Complete classes? 02

(b) State T/F and Justify: *"Dynamic programming avoids calculating the same stuff twice thus better than divide and conquers."* 02

(c) Number of ways to multiply 6 matrices is ____ and minimum number of multiplication required to find $M^6$ is _____, where size of each M is 4 x 4. 03

**Q-5 Give the answer of the following questions.**

(a) Given 10 activities along with their start and finish time as 04

$S=<A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8, A_9, A_{10}>$

$S_i=<1, 2, 3, 4, 5, 6, 7, 8, 9, 10>$

$F_i=<5, 3, 4, 6, 7, 8, 11, 10, 12, 13>$

Find the solution for the above activity scheduling problem using greedy strategy.

(b) Consider the five items along with their respective weights and values shown in below table. The knapsack has capacity, W=60; Find out the solution to the fractional knapsack. 05

| Items | $W_i$ | $V_i$ |
|---|---|---|
| $I_1$ | 5 | 30 |
| $I_2$ | 10 | 20 |
| $I_3$ | 20 | 100 |
| $I_4$ | 30 | 90 |
| $I_5$ | 40 | 160 |

(c) What is the importance of finding LCS? Determine the LCS of POLITICS and OLYMPICS using Dynamic Programming. 05

OR

**Q-5 Give the answer of the following questions.**

(a) Explain why the statement, "The running time of algorithm A is at least O ($n^2$)," is meaningless; Also mention the parameters for finding suitable algorithm among many 04 candidate algorithms. Justify parameter with suitable example.

(b) Sort the following list in increasing order of numbers **05**

9, 94, 45, 47, 28, 98, 65, 42, 78, 4, 11, 88, 6

Using each of the following methods.

a. Quick Sort

b. Merge Sort

(c) Write down an algorithm for matrix chain multiplication; Also mention its time complexity & space complexity. **05**

**Q-6 Attempt Any Two.** **14**

(a) What is the difference between 0/1 knapsack problem & fractional knapsack problem? Write a greedy algorithm for Fractional knapsack Problem and find its running time.

(b) What is the practical importance of Dijkstra's algorithm? Write down an algorithm for it.

(c) What is the use of Kruskal's algorithm? Is it greedy? Which edges form the Minimum Spanning Tree for the below given graph using Kruskal's algorithm? Consider A as a Source Node.